

5.3.16 Extended interrupt and event controller input (EXTI) characteristics

The pulse on the interrupt input must have a minimal length in order to guarantee that it is detected by the event controller.

Table 53. EXTI input characteristics

Specified by design and not tested in production.

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
PLEC	Pulse length to event controller	-	20	-	-	ns

5.3.17 XSPI interface characteristics

Unless otherwise specified, the parameters given in the following table are derived from tests performed under the ambient temperature , f_{HCLK} frequency, and V_{DD} supply voltage conditions summarized in [Table 18. General operating conditions](#) , with the following configuration:

- Output speed is set to $OSPEEDR[1:0] = 11$
- Measurement points are done at CMOS levels: $0.5 V_{DD}$
- IO compensation cell activated

Table 54. XSPI characteristics in SRD mode

Evaluated by characterization - Not tested in production.

All values apply to Octal- and Quad-SPI mode.

Delay block bypassed.

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
f_{CLK}	Clock frequency	$2.7 V < V_{DD} < 3.6 V$ $C_L = 15 pF$	-	-	72	MHz
$t_{w(CLKH)}$	OCTOSPI clock high and low time, even division	PRESCALER[7:0] = n $n = (0, 1, 3, 5, \dots, 255)$	$t_{CLK}/2 - 0.5$	-	$t_{CLK}/2 + 0.5$	ns
$t_{w(CLKL)}$			$t_{CLK}/2 - 0.5$	-	$t_{CLK}/2 + 0.5$	
$t_{w(CLKH)}$	OCTOSPI clock high and low time, odd division	PRESCALER[7:0] = n $n = (2, 4, 6, \dots, 254)$	$(n/2) \times t_{CLK} / (n+1) - 0.5$	-	$(n/2) \times t_{CLK} / (n+1)$	
$t_{w(CLKL)}$			$(n/2+1) \times t_{CLK} / (n+1) - 0.5$	-	$(n/2+1) \times t_{CLK} / (n+1)$	
$t_{s(DQ)}$	Data input setup time	-	3	-	-	
$t_{h(DQ)}$	Data input hold time		1	-	-	
$t_{v(OUT)}$	Data output valid time		-	0.5	1.5	
$t_{h(OUT)}$	Data output hold time		0	-	-	

Table 55. XSPI characteristics in DTR mode (no DQS)

Evaluated by characterization - Not tested in production.

All values apply to Octal- and Quad-SPI mode.

Delay block bypassed.

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
f_{CLK}	Clock frequency	$2.7 V < V_{DD} < 3.6 V$ $C_L = 15 pF$	-	-	72 ⁽¹⁾	MHz
$t_{w(CLKL)}$	OCTOSPI clock high and low time Even division	PRESCALER[7:0] = n $n = (0, 1, 3, 5, \dots, 255)$	$t_{CLK}/2 - 0.5$	-	$t_{CLK}/2 + 0.5$	ns
$t_{w(CLKH)}$			$t_{CLK}/2 - 0.5$	-	$t_{CLK}/2 + 0.5$	
$t_{w(CLKH)}$	OCTOSPI clock high and low time Even division	PRESCALER[7:0] = n $n = (2, 4, 6, \dots, 254)$	$(n/2) \times t_{CLK} / (n+1) - 0.5$	-	$(n/2) \times t_{CLK} / (n+1) + 0.5$	
$t_{w(CLKL)}$			$(n/2+1) \times t_{CLK} / (n+1) - 0.5$	-	$(n/2+1) \times t_{CLK} / (n+1) + 0.5$	